ASSIGNMENT-2(b) : LOOPS

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1. Add first 'n' natural numbers.

In [37]:
```
n = int(input('Enter n: '))
if n<=0:
    print('Not a natural number.')
else:
    sum = 0
    for i in range (0,n+1):
        sum += i
    print(f'Sum of first {n} natural numbers is {sum}')
```

```
Sum of first 10 natural numbers is 55
```

2. Show the Fibonacci series up to 10

In [27]:
```
a,b = 0,1
print('0 1', end=' ')
for i in range(3,11):
    c = a+b
    print(c, end=' ')
    a,b = b,c
```

```
0 1 1 2 3 5 8 13 21 34
```

3. Find the factorial of a number

In [30]:
```
n = int(input('Enter a number: '))
if n<0:
    print('Factorial of a negative number does not exist.')
else:
    f = 1
    for i in range(1,n+1):
        f *= i
    print(f'Factorial of {n} is {f}')
```

```
Factorial of 6 is 720
```

4. Find the sum of squares of first 'n' natural numbers

In [36]:
```
n = int(input('Enter a number: '))
if n<=0:
    print('Not a natural number.')
else:
    sum = 0
    for i in range(1,n+1):
        sum += i*i
    print(f'Sum of squares of first {n} natural numbers is {sum}')
```

```
Sum of squares of first 45 natural numbers is 31395
```

5. Show the multiplication table from 1 to 10

In [16]:
```python
for i in range(1,11):
    print(f'Multiplication Table of {i}: ')
    for j in range(1,11):
        print(f'{i} x {j} = {i*j}')
    print()
```

```
Multiplication Table of 1:
1 x 1 = 1
1 x 2 = 2
1 x 3 = 3
1 x 4 = 4
1 x 5 = 5
1 x 6 = 6
1 x 7 = 7
1 x 8 = 8
1 x 9 = 9
1 x 10 = 10

Multiplication Table of 2:
2 x 1 = 2
2 x 2 = 4
2 x 3 = 6
2 x 4 = 8
2 x 5 = 10
2 x 6 = 12
2 x 7 = 14
2 x 8 = 16
2 x 9 = 18
2 x 10 = 20

Multiplication Table of 3:
3 x 1 = 3
3 x 2 = 6
3 x 3 = 9
3 x 4 = 12
3 x 5 = 15
3 x 6 = 18
3 x 7 = 21
3 x 8 = 24
3 x 9 = 27
3 x 10 = 30

Multiplication Table of 4:
4 x 1 = 4
4 x 2 = 8
4 x 3 = 12
4 x 4 = 16
4 x 5 = 20
4 x 6 = 24
4 x 7 = 28
4 x 8 = 32
4 x 9 = 36
4 x 10 = 40

Multiplication Table of 5:
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50

Multiplication Table of 6:
6 x 1 = 6
```

```
6 x 2 = 12
6 x 3 = 18
6 x 4 = 24
6 x 5 = 30
6 x 6 = 36
6 x 7 = 42
6 x 8 = 48
6 x 9 = 54
6 x 10 = 60

Multiplication Table of 7:
7 x 1 = 7
7 x 2 = 14
7 x 3 = 21
7 x 4 = 28
7 x 5 = 35
7 x 6 = 42
7 x 7 = 49
7 x 8 = 56
7 x 9 = 63
7 x 10 = 70

Multiplication Table of 8:
8 x 1 = 8
8 x 2 = 16
8 x 3 = 24
8 x 4 = 32
8 x 5 = 40
8 x 6 = 48
8 x 7 = 56
8 x 8 = 64
8 x 9 = 72
8 x 10 = 80

Multiplication Table of 9:
9 x 1 = 9
9 x 2 = 18
9 x 3 = 27
9 x 4 = 36
9 x 5 = 45
9 x 6 = 54
9 x 7 = 63
9 x 8 = 72
9 x 9 = 81
9 x 10 = 90

Multiplication Table of 10:
10 x 1 = 10
10 x 2 = 20
10 x 3 = 30
10 x 4 = 40
10 x 5 = 50
10 x 6 = 60
10 x 7 = 70
10 x 8 = 80
10 x 9 = 90
10 x 10 = 100
```

6. Find the prime numbers from 2 to 10

In [38]:
```
print('Prime Numbers :', end=' ')
for i in range(2, 11):
    flag = 0
    for j in range(2,int(i**0.5 + 1)):
        if i%j==0:
            flag = 1
            break
    if flag==0:
        print(i, end=' ')
```

```
Prime Numbers : 2 3 5 7
```